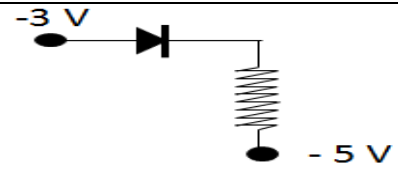
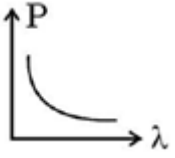
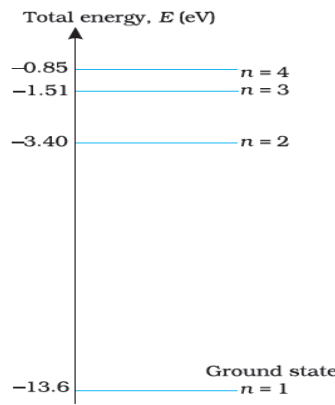
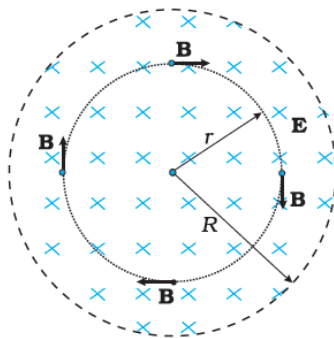
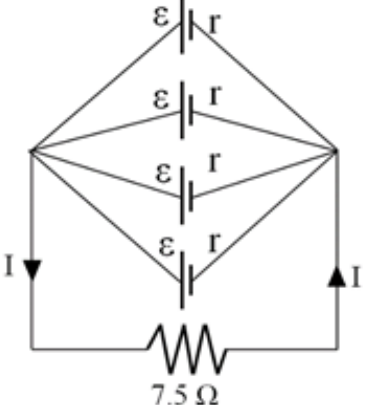
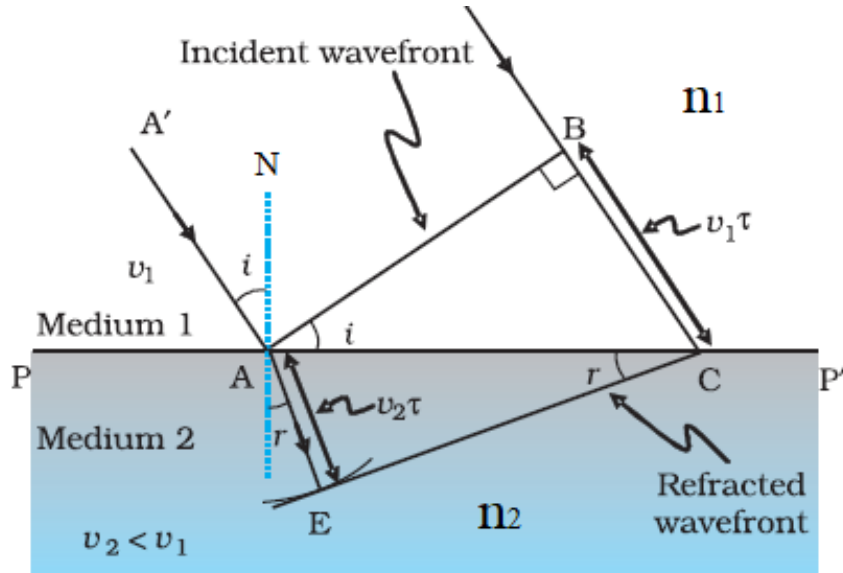
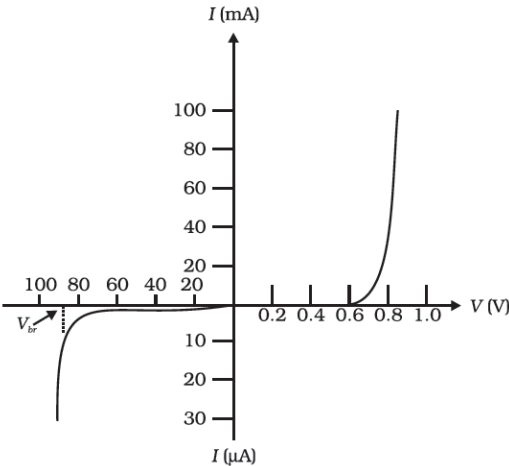
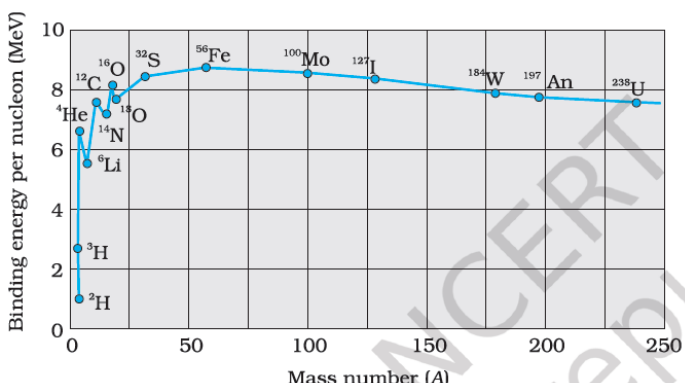
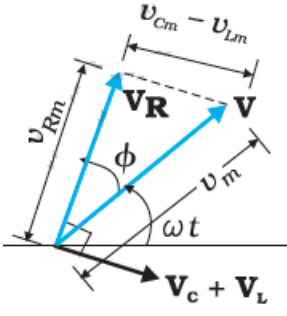


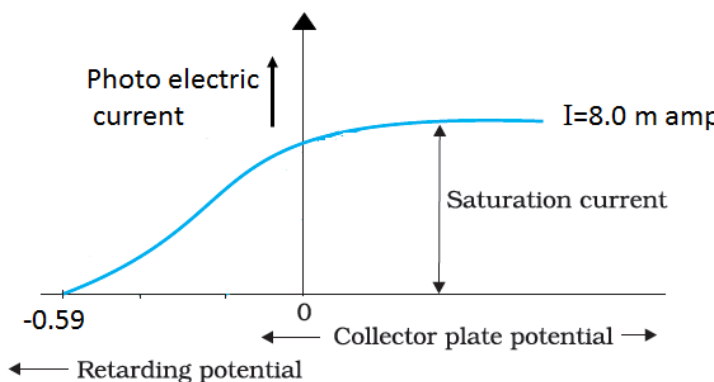
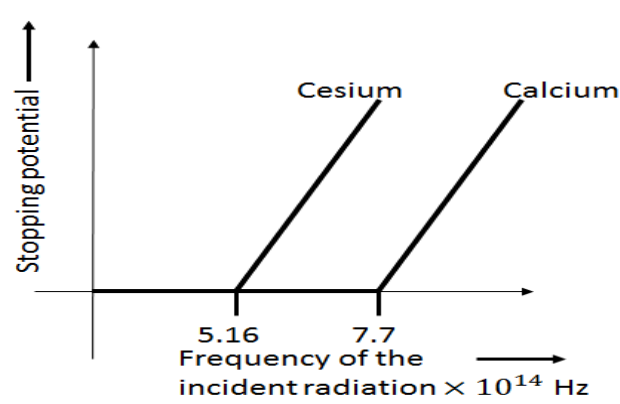
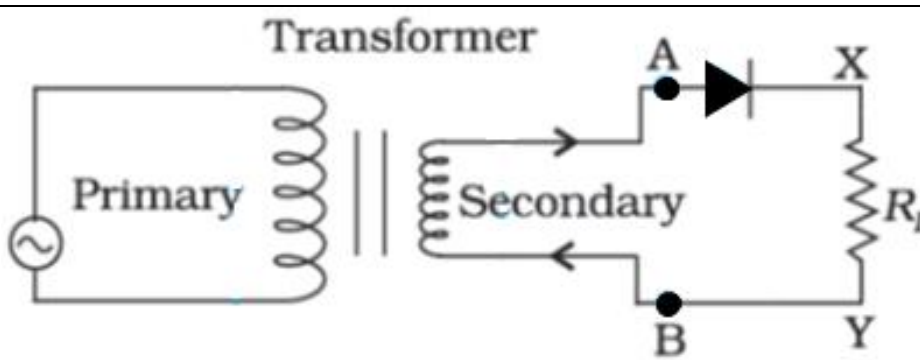
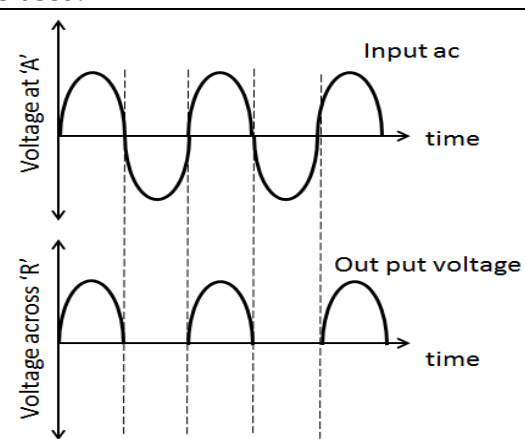
01	C)	0^0	1
02	A)	1	1
03	C)	Ammeter of range 0 -10 amp	1
04	D)	$q_3 > q_1 > q_2$	1
05	D)	4f	1
06	A)	Spherical wave front with radius f	1
07	C)	$3/4$	1
08	B)		1
09	A)	200 MeV	1
10	A)		1
11	B)	Diverging lens of focal length 60 cm	1
12	B)	Electric field $\neq 0$ and potential = 0	1
13	A)	Assertion and Reason are true and Reason is the correct explanation of Assertion.	1
14	C)	Assertion is true but Reason is false	1
15	D)	Both Assertion and Reason are false.	1
16	B)	Both Assertion and Reason are true but Reason is not the correct explanation of Assertion	1
SECTION ‘C’			
17	Stable equilibrium when $\vec{m}$ is parallel to $\vec{B}$ Corresponding potential energy $U = -mB = -4.8 \times 10^{-2} \text{ joule}$ Stable equilibrium when $\vec{m}$ is anti-parallel to $\vec{B}$ Corresponding potential energy $U = +mB = +4.8 \times 10^{-2} \text{ joule}$		2
18	Electrical potential on the surface of the spherical shell $V = \frac{1}{4\mu\epsilon_0} \frac{q}{r}$ $V = \frac{9 \times 10^9 \times 6 \times 10^{-6}}{0.2} = 2.7 \times 10^5 \text{ volt}$ Since electric field inside the hollow sphere is zero, hence V remains constant throughout the volume. Potential at the centre of the sphere is $2.7 \times 10^5 \text{ volt}$		2

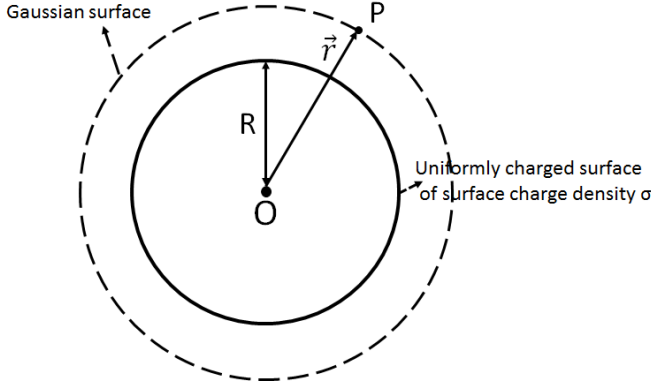
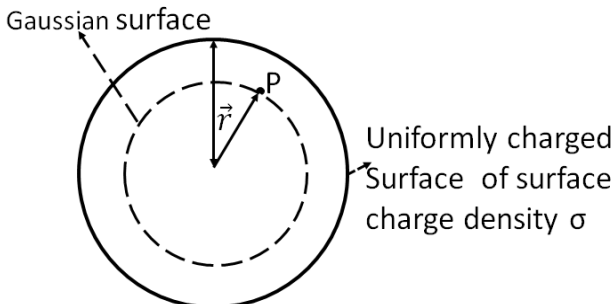
19	I	i.	$r_n = n^2 r_1 = 3^2 \times 5.3 \times 10^{-11}$ $2\pi r_n = n\lambda_n$ $\lambda_n = \frac{2\pi r_n}{n}$ $\lambda_n = \frac{2\pi n^2 r_1}{n} = 2\pi n r_1$ $\lambda_3 = \frac{2 \times 3.14 \times 3^2 \times 5.3 \times 10^{-11}}{4}$ $= 37.44 \times 10^{-11} \text{ meter}$	$\frac{1}{2}$	2
	[OR]				
	II	a)		1	
			$E = -13.6 \text{ eV}$ $U = 2E = -27.2 \text{ eV}$ $K = -E = 13.6 \text{ eV}$	$\frac{1}{2}$ $\frac{1}{2}$	
20	a)		1	2	
		Correct Statement or $\int \vec{B} \cdot d\vec{l} = \mu_0 i_c + \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$	1		
21		In the low current region, the current generate less amount of heat which is dissipated to surrounding and the conductor is at thermal equilibrium with the surrounding so there is no rise in temperature hence the resistivity of the material is not changing. At high current region heat generated is more than the heat dissipated. The heat retained by the conductor increases the temperature of the wire hence the resistivity increases in high current region.	1 1	2	

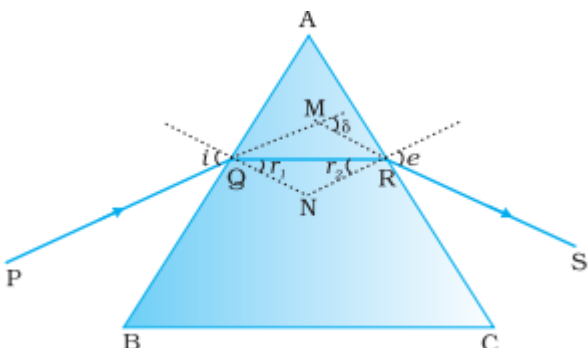
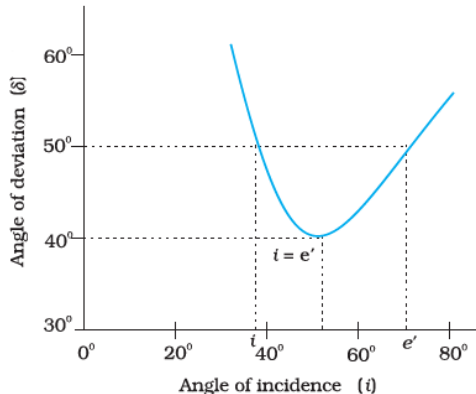
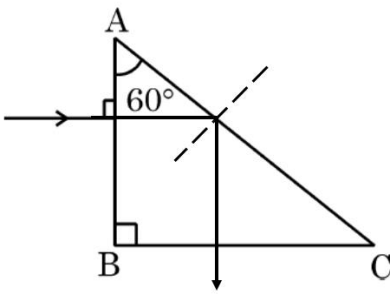
		SECTION 'C'		
22	I		1	3
		Let r' be the net internal resistance of the group of four cells then $r' = R \left(\frac{\varepsilon - V}{V} \right)$ $= 7.5 \left(\frac{2 - 1.6}{1.6} \right) = \frac{15}{8} \Omega$ As 'r' is the internal resistance of each cell $\frac{1}{r'} = \frac{4}{r}$ $\therefore r = 4 \times r'$ $= 4 \times \frac{15}{8} = 7.5 \Omega$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	
23	a)		1	3
		In the right angled triangle ABC $\sin i = \frac{BC}{AC} = \frac{\tau v_1}{AC}$ In the right angled triangle ACE $\sin r = \frac{AE}{AC} = \frac{\tau v_2}{AC}$ $\frac{\sin i}{\sin r} = \frac{\tau v_1}{\tau v_2} = \frac{v_1}{v_2} = \frac{n_2}{n_1} = n_{21}$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	
	b) frequency		$\frac{1}{2}$	

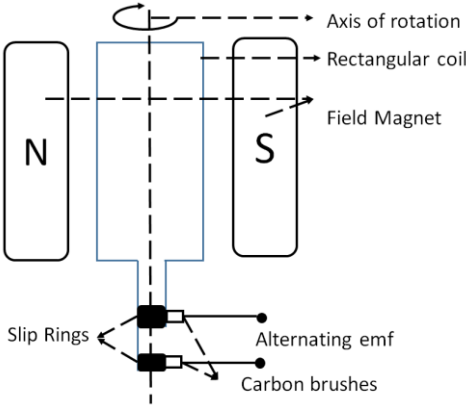
24			1	3
	In forward biased condition, the current first increases very slowly, almost negligibly, till the voltage across the diode crosses a certain value. After the characteristic voltage, the diode current increases significantly. This voltage is called the threshold voltage or cut-in voltage. Reverse current is due to the minority carriers of both sides of the diode. Very small reverse voltage is sufficient to sweep all the minority carriers from one side of the junction to the other side of the junction.		1 ½ ½	
25			1½	3
	$E_b = \text{mass defect in } amu \times 931.5 \text{ MeV}/c^2$ $E_{bn} = \frac{E_b}{A}$ $E_{bn} = \frac{0.13691 \times 931.5}{16} = \frac{127.5}{16} = 7.96 \text{ MeV}/c^2$		½ ½ ½.	
26	I	a) Flux linkage through a solenoid of N turns is proportional to the current through the solenoid $N\Phi_B \propto I$ $N\Phi_B = LI$, where L is the self inductance of the solenoid Consider a solenoid of cross sectional area A and length l, having n turns per unit length. The magnetic field due to a current I flowing in the solenoid is $B = \mu_0 n I$ The total flux linked with the solenoid is $N\Phi_B = (nl)(\mu_0 nl)(A)$ $N\Phi_B = \mu_0 n^2 AlI$ $LI = \mu_0 n^2 AlI$ $L = \mu_0 n^2 Al$	½ ½ ½ ½	3

		$L = \mu_0 \mu_r n^2 A l$	1/2	
		b) The self-induced emf opposes any change in the current in the circuit (OR) consequence of Lenz's law.	1/2	
		[OR]		
	II	X - Resistor Y - Inductor Z - Capacitor	1/2 1/2 1/2	
			1/2	
		From the phasor diagram $v_m^2 = v_{Rm}^2 + (v_{Cm} + v_{Lm})^2$ <i>Substituting the values of v_{Rm}, v_{Cm} and v_{Lm}</i> $v_m^2 = (i_m R)^2 + (i_m X_c + i_m X_L)^2$ $i_m = \frac{v_m}{(R^2 + [X_c - X_L])^{\frac{1}{2}}}$ By analogy to the resistance in a circuit, we introduce the impedance Z in an ac circuit $i_m = \frac{v_m}{Z}$ $Z = (R^2 + [X_c - X_L])^{\frac{1}{2}}$	1/2 1/2	
27	a)	Stopping potential for cesium $eV_0 = h\nu - h\nu_0$ $V_0 = \frac{6.63 \times 10^{-34} (6.6 - 5.16) \times 10^{14}}{1.6 \times 10^{-19}} = 0.59 \text{ vol}$ Saturation current $I = N \times e = 50 \times 10^{15} \times 1.6 \times 10^{-19} = 8 \text{ mA}$	1/2 1/2 1/2	3

					$\frac{1}{2}$	
	b) 				$\frac{1}{2}$ + $\frac{1}{2}$	
28					1	3
	If an alternating voltage is applied across a diode in series with a load, a pulsating voltage will appear across the load only during the half cycles of the ac input during which the diode is forward biased.				1	
					1	
SECTION 'D'						
29	I	A)	$2\pi / \tan \theta$		1	4
	II	C)	Frequency		1	
	II	B)	Circular motion in x-y plane		1	
	IV	i.	C)	2R	1	

		[OR]								
		ii.	D)	Zero						
30.	I	B)	Yes they should have different refractive index.						1	4
	II	D)	all the above						1	
	II	A)	then the refractive index of the liquid is equal to 1.47						1	
	IV	i	C)	Doubles	Same	Same	Becomes half	1		
		[OR]								
		ii.	B)	3.9 f						
SECTION 'E'										
31	I	a)	When $r > R$ 						1/2	5
			The electric field at each point of the Gaussian surface, therefore, has the same magnitude E and is along the radius vector at each point. Thus, $\vec{E}$ and $\vec{ds}$ at every point are parallel and the flux through each element is $E ds$. Summing over all ds, the flux through the Gaussian surface is $E \times 4 \pi r^2$. The charge enclosed is $\sigma \times 4 \pi R^2$ By Gauss's law flux through the Gaussian surface is $\frac{\sigma \times 4 \pi R^2}{\epsilon_0}$ $\therefore E \times 4 \pi r^2 = \frac{\sigma \times 4 \pi R^2}{\epsilon_0}$ $E = \frac{\sigma R^2}{\epsilon_0 r^2} = \frac{q}{4 \pi \epsilon_0 r^2}$						1/2	
			When $r < R$ 						1/2	
			Gaussian surface does not enclose any charge so field inside the charged surface is zero						1/2	

32	I	a)	i. 	1/2	5
			In the quadrilateral AQNR $\angle A + \angle QNR = 180^\circ$ In the triangle QNR $r_1 + r_2 = \angle A$ δ is the exterior angle of the triangle MQR $\delta = e - r_2 + i - r_1$ $\delta = e + i - A$	1/2 1/2 1/2	
			ii. 	1	
		b)		1/2	
			The incident ray makes normal incident on the first interface AB and travel un-deviated parallel to the base. The angle of incident on the second interface AC is 60° $\sin i_c = \frac{1}{\sqrt{2}}$ $i_c = 45^\circ$ At the second interface the light undergoes total internal reflection. The reflected ray makes normal incidence on the base of the prism travels back to the first medium with a deviation of 90°.	1/2 1/2 1/2	
		[OR]			
	II	a)	$Y_1 = a \cos \omega t$ $Y_2 = a \cos(\omega t + \phi)$ $Y = a \cos \omega t + a \cos(\omega t + \phi)$ $Y = 2a \cos \frac{\phi}{2} \cos \left(\omega t + \frac{\phi}{2} \right)$	1/2	

			$Y = A \cos\left(\omega t + \frac{\phi}{2}\right)$ $A = 2a \cos \frac{\phi}{2}, \text{ the resultant amplitude}$ i. $I = 4I_0 \cos^2 \frac{\phi}{2}$ ii. $I = 4I_0 \frac{1}{2} = 2I_0$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	
		b)	Light waves coming out from two independent sources of light will not have any fixed phase relationship and would be incoherent	1	
		c)	When the path difference = λ and I_0 is the intensity of the each interfering waves then the intensity of light at a point on the screen is $k = 4I_0$ When path difference = $\lambda/3$ phase difference is 120° Hence the intensity of light at a point where path difference $\lambda/3$ is $I = k \cos^2 \frac{120}{2}$ $I = k \cos^2 60 = \frac{k}{4}$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	
33	I	a)		1	5
		Working:- The coil is mechanically rotated in the uniform magnetic field by some external means. The rotation of the coil causes the magnetic flux through it to change, so an emf is induced in the coil. The ends of the coil are connected to an external circuit by means of slip rings and brushes. The coil is mechanically rotated in the uniform magnetic field by some external means. The rotation of the coil causes the magnetic flux through it to change, so an emf is induced in the coil. The ends of the coil are connected to an external circuit by means of slip rings and brushes. When the coil is rotated with a constant angular speed ω, the angle θ between the magnetic field vector $\vec{B}$ and the area vector $\vec{A}$ of the coil at any instant t is $\theta = \omega t$. As a result, the effective area of the coil exposed to the magnetic field lines changes with time. The flux linked with the coil of N turns at any given time is $\Phi_B = NAB \cos \omega t$ Differentiating both sides with respect to time and using Faradays laws $\text{instantaneous emf } \varepsilon = NAB\omega \sin \omega t$ $\varepsilon = \varepsilon_0 \sin \omega t$		$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	

